

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: BTech

Branch: Electrical Engineering and Allied

Semester : VI

Subject Code & Name: BTEEC601 & Switchgear and Protection

Max Marks: 60

Date: 18/12/2024

Duration: 3 Hr

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	HRC stands for		1				
	<table border="1"> <tr> <td>a. High Rupturing capacity Fuse</td> <td>b. High rated current fuse</td> <td>c. High reactive current fuse</td> <td>d. High ratio fuse</td> </tr> </table>	a. High Rupturing capacity Fuse	b. High rated current fuse	c. High reactive current fuse	d. High ratio fuse		
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2	Purpose of backup protection is						
	<table border="1"> <tr> <td>a. To increase the speed</td> <td>b. To increase a reach</td> <td>c. To leave no blind spot</td> <td>d. To guard against failure of primary</td> </tr> </table>	a. To increase the speed	b. To increase a reach	c. To leave no blind spot	d. To guard against failure of primary		
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3	For the protection of stator winding of an alternate against internal fault involving ground the relay used is a		1				
	<table border="1"> <tr> <td>a. Biased differential relay</td> <td>b. Directional over-current relay</td> <td>c. Plain impedance relay</td> <td>d. Buchholz relay</td> </tr> </table>	a. Biased differential relay	b. Directional over-current relay	c. Plain impedance relay	d. Buchholz relay		
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4	If the fault occurs near the impedance relay, the V/I ratio will be		1				
	<table border="1"> <tr> <td>a. constant for all distances</td> <td>b. lower than that of if fault occurs away from the relay</td> <td>c. higher than that of if fault occurs away from the relay</td> <td>d. none of the above</td> </tr> </table>	a. constant for all distances	b. lower than that of if fault occurs away from the relay	c. higher than that of if fault occurs away from the relay	d. none of the above		
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5	The rating of a circuit breaker is generally determined on the basis of						
	<table border="1"> <tr> <td>a. line -to-line fault current</td> <td>b. symmetrical fault current</td> <td>c. single line to ground fault current</td> <td>d. double line to ground fault current</td> </tr> </table>	a. line -to-line fault current	b. symmetrical fault current	c. single line to ground fault current	d. double line to ground fault current		
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6	Arc in a circuit breaker is interrupted at		1				
	<table border="1"> <tr> <td>a. zero current</td> <td>b. maximum current</td> <td>c. maximum voltage</td> <td>d. maximum voltage</td> </tr> </table>	a. zero current	b. maximum current	c. maximum voltage	d. maximum voltage		
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7	Admittance relay is _____ relay					1
	a.impedance	b. directional	c. non-directional	d. none of the above		
8	The voltage appearing across the contacts after opening of the circuit breaker is called					1
	a. recovery	b. surge	c. Operating	d. none of the above		
9	Which of the following medium is employed for extinction of arc in air circuit breaker					1
	a. Water	b. Oil	c. Air	d. SF ₆		
10	A differential relay measures the vector difference between					1
	a. two currents	b. two voltages	c. two or more similar electrical quantities	d. none of the above		
11	An efficient and a well-designed protective relaying should have					1
	a. good selectivity and reliability	b. economy and simplicity	c. high speed and selectivity	d. all of the above		
12	Distance relays are generally					1
	a. split-phase relays	b. reactance relays	c. impedance relays	d. none of the above		
Q. 2 Solve the following.						
A)	Explain SF6 circuit breaker					6
B)	What are advantages and disadvantages of static relays					6
Q.3 Solve the following.						12
A)	Explain the desirable characteristics of protection system					6
B)	Explain principle of working of induction cup type relay					6
Q. 4 Solve Any Two of the following.						
A)	With neat diagram explain percentage differential protection					6
B)	Explain with neat diagram over current protection of transformer					6
C)	Name the different types of fuses and explain various application of HRC fuses					6

Q.5	Solve Any Two of the following.		12
A)	Explain Buchholz relay with neat diagram		6
B)	Explain protection of ring main and parallel feeders system		6
C)	Explain minimum oil circuit breaker		6
Q. 6	Solve Any Two of the following.		12
A)	Explain the working principle of differential relays.		6
B)	Discuss primary and backup protection system		6
C)	What are advantages and disadvantages of static relays		6
*** End ***			